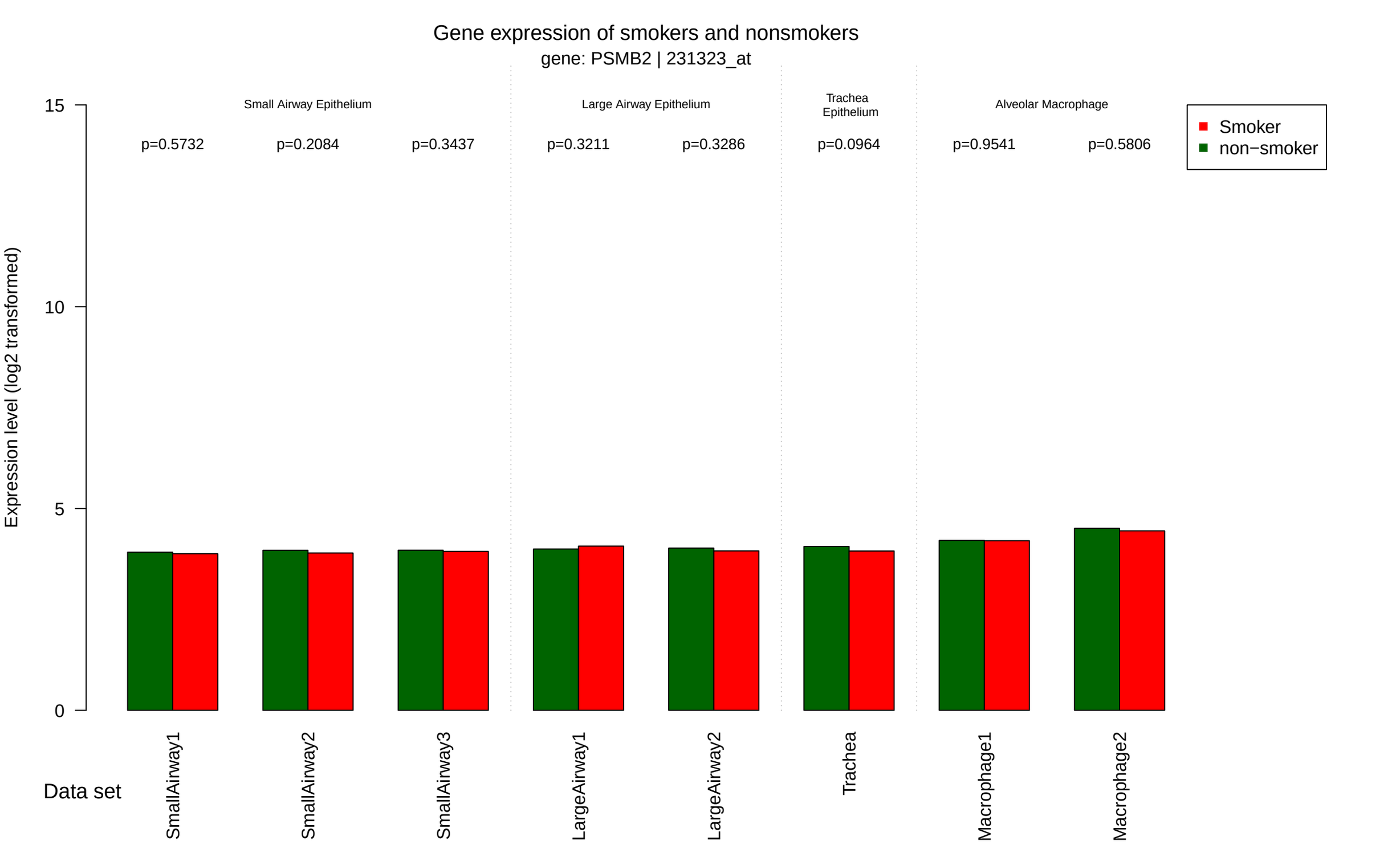
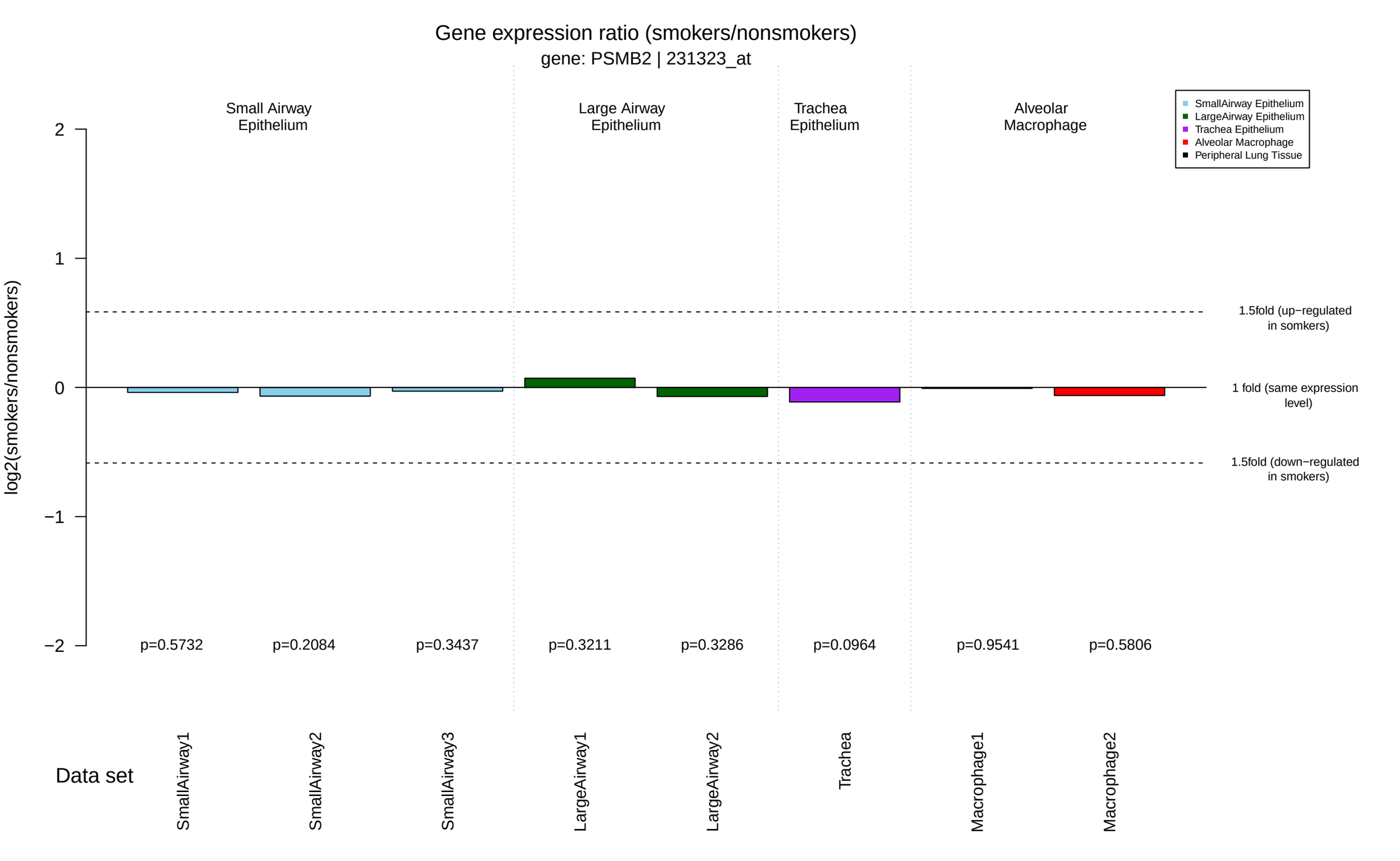
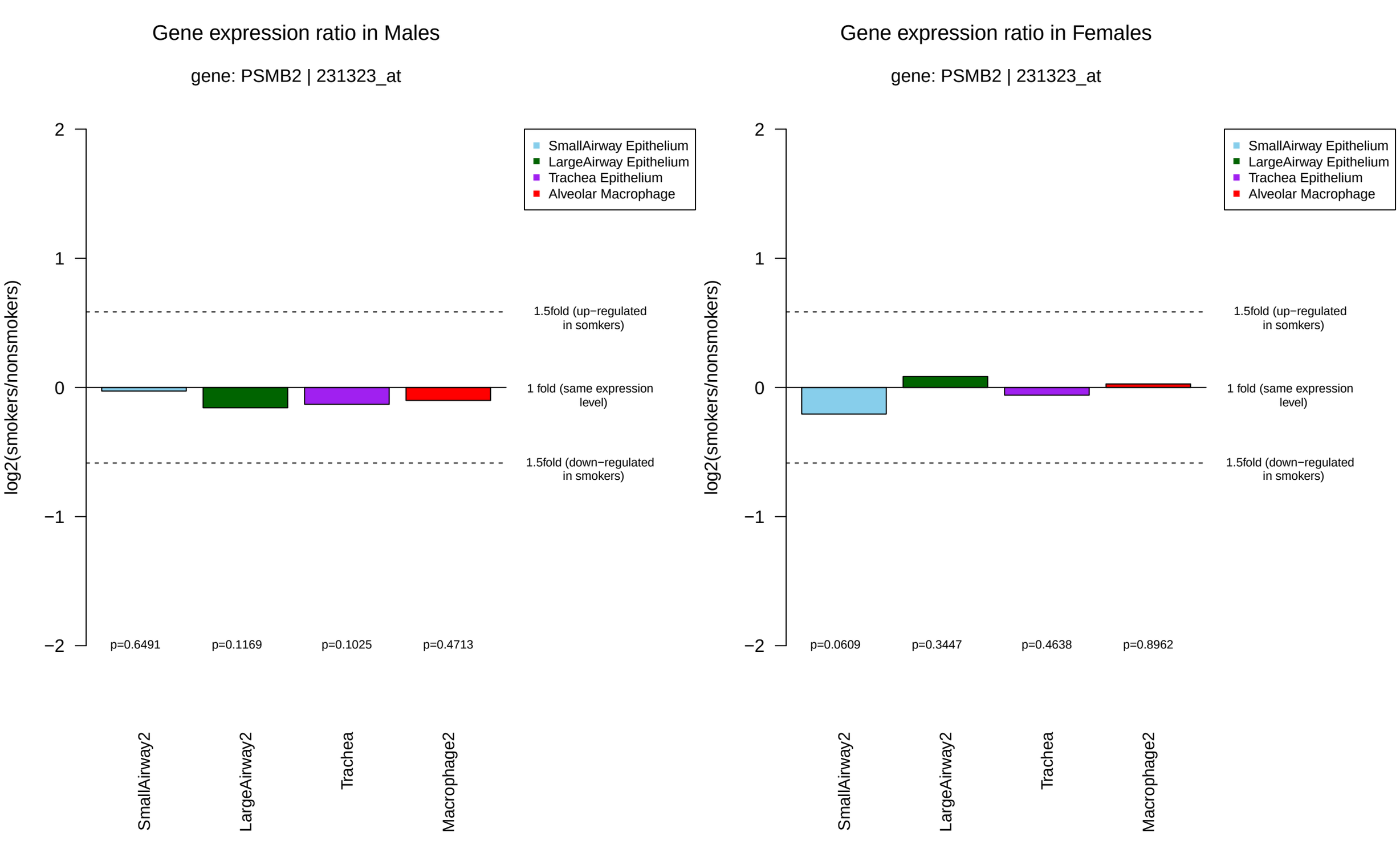
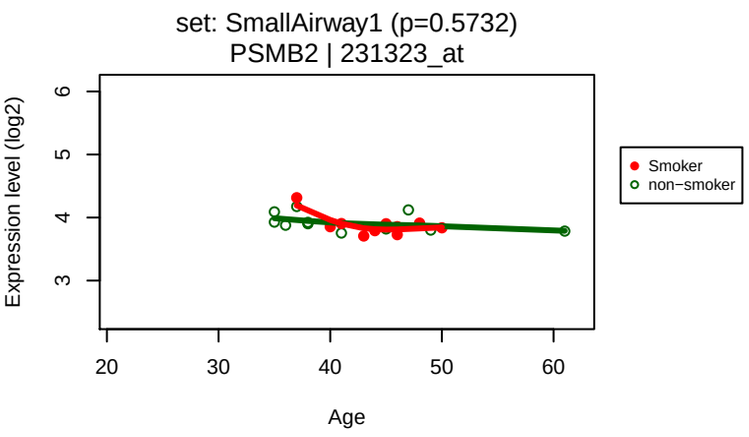
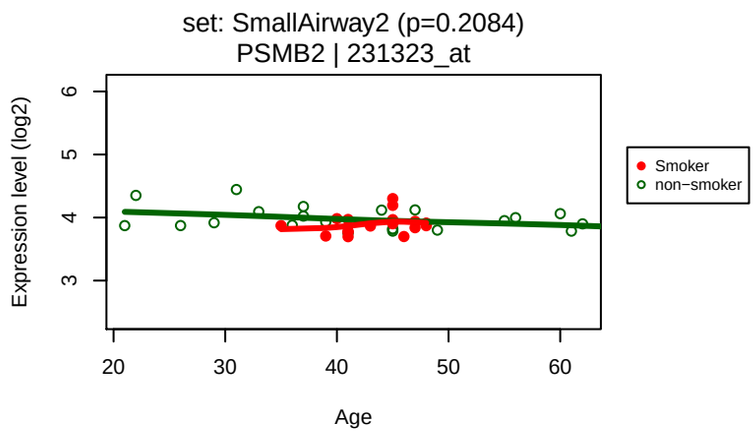
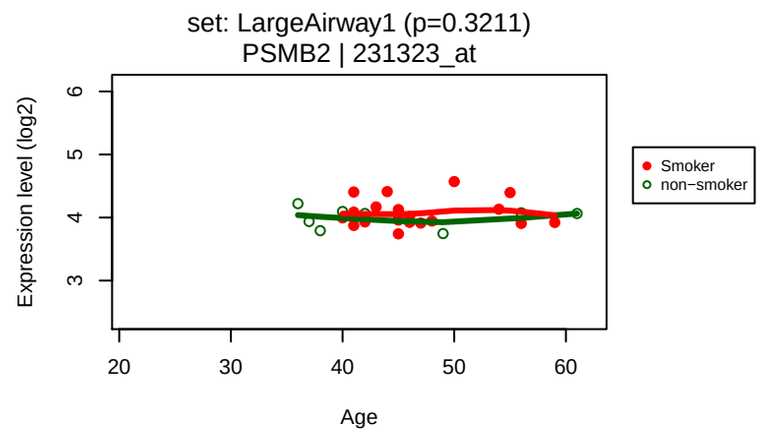

Gene expression of smokers and nonsmokers
gene: PSMB2 | 231323_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5732
p=0.2084
p=0.3437
p=0.3211
p=0.3286
p=0.0964
p=0.9541
p=0.5806
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

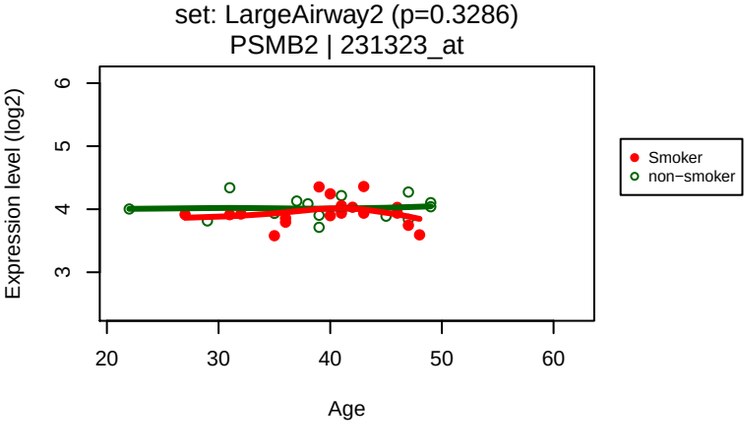

Gene expression ratio (smokers/nonsmokers)
gene: PSMB2 | 231323_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5732
p=0.2084
p=0.3437
p=0.3211
p=0.3286
p=0.0964
p=0.9541
p=0.5806
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

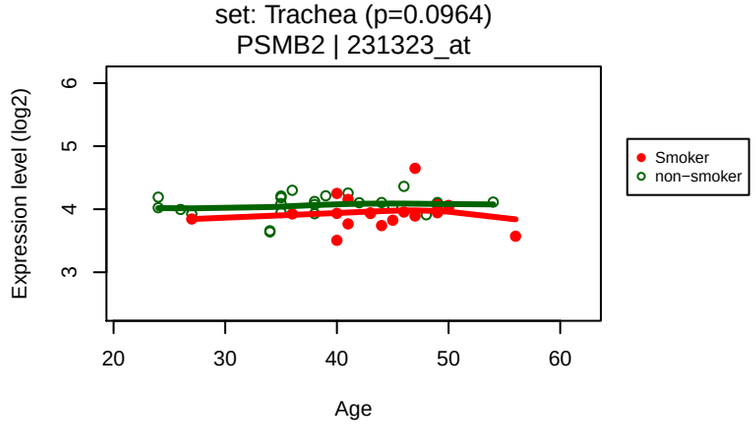

Gene expression ratio in Males
gene: PSMB2 | 231323_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6491
p=0.1169
p=0.1025
p=0.4713
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PSMB2 | 231323_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0609
p=0.3447
p=0.4638
p=0.8962
SmallAirway2
LargeAirway2
Trachea
Macrophage2

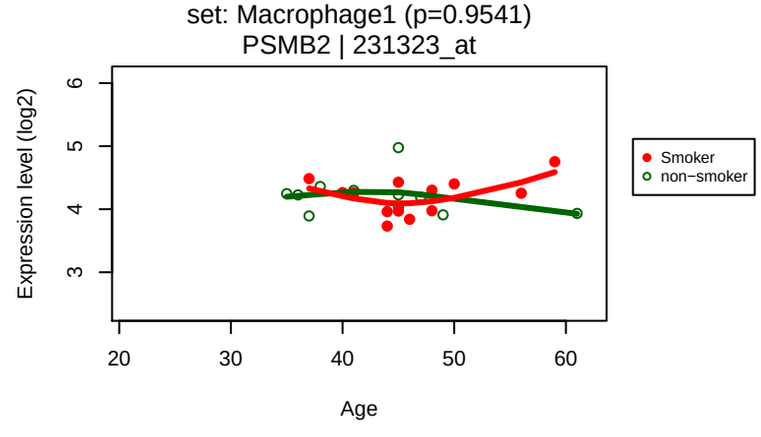

set: SmallAirway1 (p=0.5732)
PSMB2 | 231323_at
Expression level (log2)
Age
Smoker
non−smoker

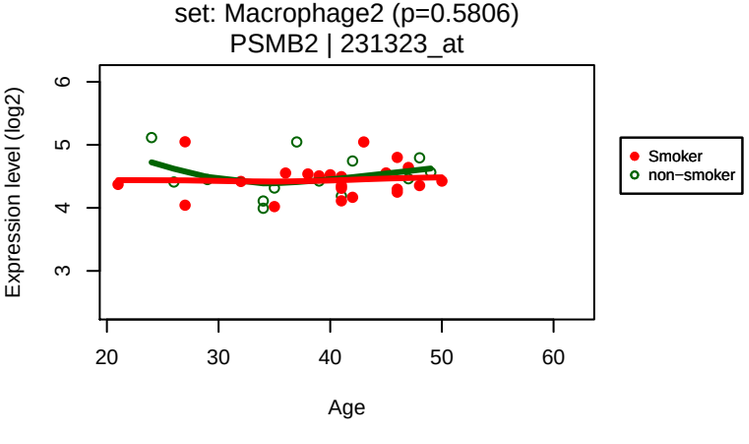

set: SmallAirway2 (p=0.2084)
PSMB2 | 231323_at
Expression level (log2)
Age
Smoker
non−smoker

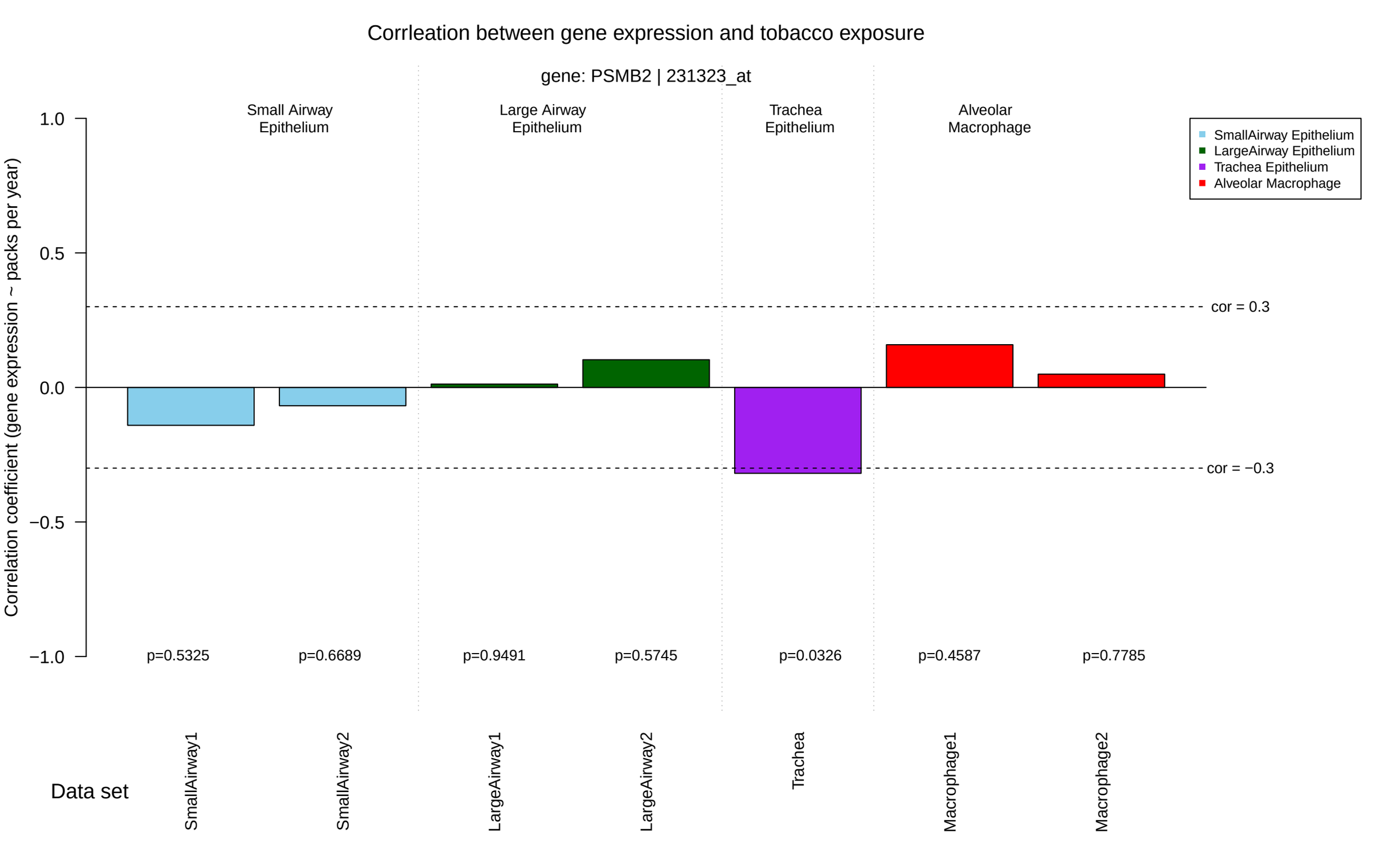

set: LargeAirway1 (p=0.3211)
PSMB2 | 231323_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3286)
PSMB2 | 231323_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0964)
PSMB2 | 231323_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9541)
PSMB2 | 231323_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5806)
PSMB2 | 231323_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PSMB2 | 231323_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5325
p=0.6689
p=0.9491
p=0.5745
p=0.0326
p=0.4587
p=0.7785
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2